

General Theory of Relativity II: Extended Adapted Coordinates and the Standard Table

Problem Set 4

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Topic

Brovary Optical Calibration

Submission

Submit one PDF. Show the derivation, identify every approximation, and include a short consistency check. The maximum file size is 25 MB.

Problems

1. Derive the first displayed relation used in Brovary Optical Calibration. State all assumptions and conventions.

$$\delta\theta(r) = \gamma \, 2GM/(c^2 r)$$

2. Evaluate a nontrivial case using the second relation. Carry units and uncertainty where applicable.

$$\gamma = 2.00 \pm 0.22$$

3. Compare the third relation in two limiting regimes and explain the physical meaning of the result.

$$c_{read} = \lambda c_{medium}$$

Show enough intermediate work that another student can reproduce the calculation.